

(1) *The global section functor*

$$\Gamma : \text{Mod}_{\underline{\mathcal{S}}_X} \rightarrow \text{Mod}_{\underline{\mathcal{S}}_X(X)}$$

is exact and establishes an equivalence of categories between the category of sheaves of right $\underline{\mathcal{S}}_X$ -modules and the category of right modules over the global sections $\underline{\mathcal{S}}_X(X)$ of $\underline{\mathcal{S}}_X$.

(2) *If $\underline{E} \in \text{Mod}_{\underline{\mathcal{S}}_X}$ locally has finite resolutions by finitely generated free $\underline{\mathcal{S}}_X$ -modules, then $\Gamma(X, \underline{E})$ has a finite resolution by finitely generated projective modules.*

(3) *The derived category of perfect complexes of sheaves $\text{D}_{\text{Perf}}(\text{Mod}_{\underline{\mathcal{S}}_X})$ is equivalent to the derived category of perfect complexes of modules $\text{D}_{\text{Perf}}(\text{Mod}_{\underline{\mathcal{S}}_X(X)})$.*

By this theorem, there is a (strict) perfect complex of A^0 -modules $(E, \mathbb{E}^0)$ and a quasi-isomorphism $e^0 : (E^\bullet, \mathbb{E}^0) \rightarrow (F^\bullet, \mathbb{F}^0) = (\Gamma(M, \underline{\mathcal{C}}_F^\infty), D)$. We shall follow the argument of Theorem 3.2.7 of [Blo05] to construct the higher components $\mathbb{E}^i$ of $\mathbb{Z}$ -connection along with the higher components of a morphism e^i .

On $F^\bullet$, we have a $\mathbb{Z}$ -connection

$$\mathbb{F} = D \otimes 1 + 1 \otimes d : F^\bullet \rightarrow F^\bullet \otimes_{A^0} A^1$$

. The idea is to transfer this $\mathbb{Z}$ -connection to $E^\bullet$ which is compatible with the quasi-isomorphism on H^0 's. Note that we have an induced connection

$$\mathbb{H}^k : H^k(F^\bullet, \mathbb{F}^0) \rightarrow H^k(F^\bullet, \mathbb{F}^0) \otimes_{A^0} A^1$$

for each k . First we will transfer this connection to a connection on $H^k(E^\bullet, \mathbb{E}^0)$, and we have the following commutative diagram

$$\begin{array}{ccc} H^k(E^\bullet, \mathbb{E}^0) & \xrightarrow{\mathbb{H}^k} & H^k(E^\bullet, \mathbb{E}^0) \otimes_{A^0} A^1 \\ \downarrow e^0 & & \downarrow e^0 \otimes 1 \\ H^k(F^\bullet, \mathbb{F}^0) & \xrightarrow{\mathbb{H}^k} & H^k(F^\bullet, \mathbb{F}^0) \otimes_{A^0} A^1 \end{array}$$

Note that $e^0 \otimes 1$ is a quasi-isomorphism since A^1 is flat over A^0 . We need the following lemma.

Lemma 2.10. *Given a bounded complex of finitely generated projective A^0 -modules $(E^\bullet, \mathbb{E}^0)$ with connections $\mathbb{H}^k : H^k(F^\bullet, \mathbb{F}^0) \rightarrow H^k(F^\bullet, \mathbb{F}^0) \otimes_{A^0} A^1$ for each k , there exists connections*

$$\tilde{\mathbb{H}}^k : E^k \rightarrow E^k \otimes_{A^0} A^1$$

lifting $\mathbb{H}^k$, i.e.

$$\tilde{\mathbb{H}}^k \mathbb{E}^0 = (\mathbb{E}^0 \otimes 1) \tilde{\mathbb{H}}^k$$

for each k and the connection induced on the cohomology is $\mathbb{H}^k$.

Proof. This is Lemma 3.2.8 in [Blo05] and Lemma 4.6 in [BS14]. Since $E^\bullet$ is bounded, let $[N, M]$ be its magnitude. Pick some arbitrary connection ∇ on E^M . Consider the following diagram whose rows are exact